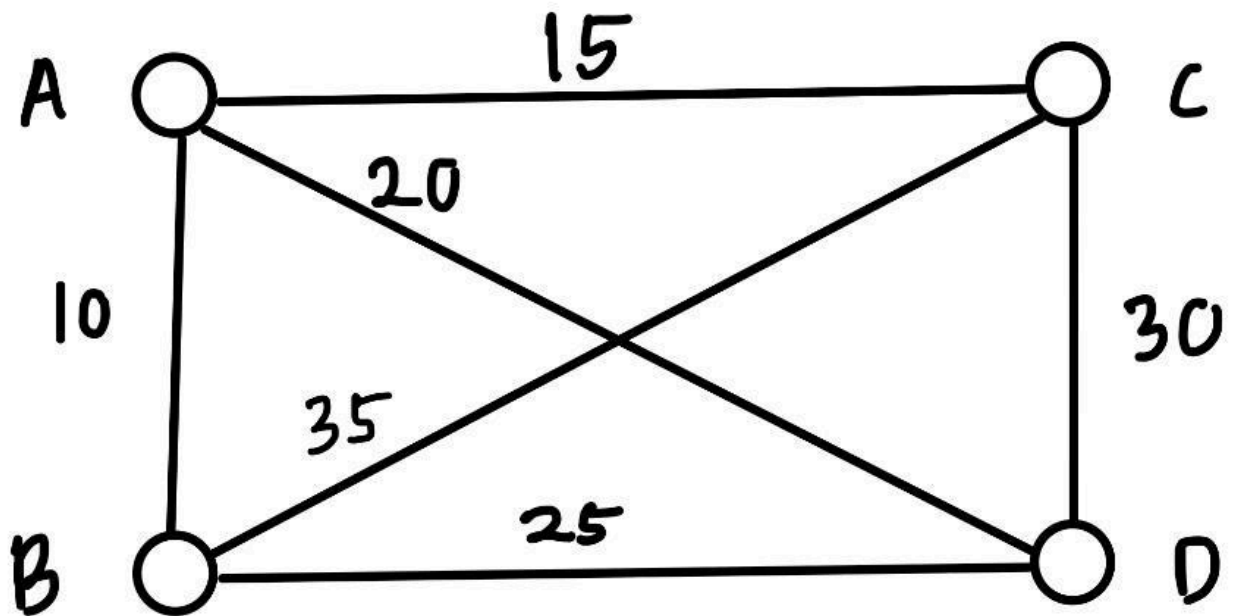


- A. Backtracking solves the Travelling Salesman Problem by exploring all possible sequences of cities recursively. Starting from a city, it visits unvisited cities one by one, keeping track of the total distance. Once all cities are visited, it returns to the starting city and updates the minimum distance if needed. It then backtracks to explore other paths. Why appropriate: It guarantees finding the optimal solution by systematically checking all routes and can prune paths that exceed the current minimum distance, making it suitable for exact solutions in small to medium-sized TSP instances.



c)	A-B-C-D-A	$10 + 35 + 30 + 20 = 95$
	A-B-D-C-A	$10 + 25 + 30 + 15 = 80$
	A-C-B-D-A	$15 + 35 + 25 + 20 = 95$
	A-C-D-B-A	$15 + 30 + 25 + 10 = 80$
	A-D-B-C-A	$20 + 25 + 35 + 15 = 95$
	A-D-C-B-A	$20 + 30 + 35 + 10 = 95$
Optimal distance tour:		
A → B → D → C → A or A → C → D → B → A		
Minimum Total Distance: 80		